Outline

1. Summary
2. Vector space $\mathbb{R}^n$
3. Limits
4. Continuity
5. Partial derivatives
6. The Chain Rules
7. Approximations

5. The Chain Rules

- Recall for functions of one variable.

The Chain Rule for functions of one variable gives the way for differentiating a composite function: if $y = f(x)$ and $x = g(t)$ are two differentiable functions, then

$$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}.$$

For functions of two or more variables, the Chain Rule has several versions, which depend on the number of variables of the given functions.

• The Chain Rule - Case 1

- For a function $f(x, y)$, when both x and y are functions of ONE variable.

Theorem (Chain Rule for $f(x, y)$: Case 1)

If $w = f(x, y)$ is differentiable and $x = x(t)$, $y = y(t)$ are differentiable functions of t, then the composite function $w = f\big(x(t), y(t)\big)$ is also differentiable w.r.t. t and

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt},$$

that is

$$\frac{dw}{dt} = f_x\big(x(t), y(t)\big)x'(t) + f_y\big(x(t), y(t)\big)y'(t).$$

Example

Suppose $w = x^2y + 3xy^4$, where $x = e^t$ and $y = \sin t$.

(a) Find $\dfrac{dw}{dt}$.

(b) What is the rate of change in w at $t = 0$?

Solution.

(a) We have

$$\begin{aligned}\frac{dw}{dt} &= \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} \\ &= (2xy + 3y^4)e^t + (x^2 + 12xy^3)\cos t \qquad (*) \\ &\text{(substitute } x = e^t,\ y = \sin t) \\ &= (2e^t \sin t + 3\sin^4 t)e^t + (e^{2t} + 12e^t \sin^3 t)\cos t. \qquad (**)\end{aligned}$$

(b) Since the rate of change in w at $t = 0$ is $\left.\frac{dw}{dt}\right|_{t=0}$, we use $(**)$

$$\frac{dw}{dt} = (2e^t \sin t + 3\sin^4 t)e^t + (e^{2t} + 12e^t \sin^3 t)\cos t$$

to get

$$\left.\frac{dw}{dt}\right|_{t=0} = (2e^0 \sin 0 + 3\sin^4 0)e^0 + (e^{2\cdot 0} + 12e^0 \sin^3 0)\cos 0$$

(substitute $e^0 = 1$, $\sin 0 = 0$, $\cos 0 = 1$) $= 1.$

- **Remark**. We can also use $(*)$

$$\frac{dw}{dt} = (2xy + 3y^4)e^t + (x^2 + 12xy^3)\cos t$$

to evaluate values of x and y at $t = 0$, and then $\left.\frac{dw}{dt}\right|_{t=0}$.

Example

The radius of a right circular cone is increasing at a rate of 1.8cm/s, while its height is decreasing at a rate of 2.5cm/s. At what rate is the volume of the cone changing when the radius is 24cm and the height is 14cm?

Solution. Volume V (cm^3) of a right circular cone with radius r (cm) and height h (cm) is

$$V = \frac{1}{3}\pi r^2 h$$

(as a function of r and h).

Also by the assumptions, we have

$$\frac{dr}{dt} = 1.8, \quad \text{and} \quad \frac{dh}{dt} = -2.5$$

Example (cont.)

From $V = \frac{1}{3}\pi r^2 h$, applying the Change Rule, we have

$$\begin{aligned} \frac{dV}{dt} &= \frac{\partial V}{\partial r} \cdot \frac{dr}{dt} + \frac{\partial V}{\partial h} \cdot \frac{dh}{dt} \\ &= \frac{\pi}{3}\Big(2rh \cdot \underbrace{\frac{dr}{dt}}_{1.8} + r^2 \cdot \underbrace{\frac{dh}{dt}}_{-2.5}\Big). \end{aligned}$$

Hence, with $r = 24$ and $h = 14$, we obtain

$$\frac{dV}{dt} = \frac{\pi}{3}\Big(2 \cdot 24 \cdot 14 \cdot (1.8) + (24)^2 \cdot (-2.5)\Big) = -76.8\pi \ (\text{cm}^3/s).$$

- For a function $f(x, y, z)$, when x, y, and z are functions of ONE variable.

Theorem (Chain Rule for $f(x, y, z)$: Case 1)

If $w = f(x, y, z)$ is differentiable and $x = x(t)$, $y = y(t)$, and $z = z(t)$ are differentiable functions of t, then the composite function $w = f\big(x(t), y(t), z(t)\big)$ is also differentiable w.r.t t and

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt},$$

that is

$$\begin{aligned}\frac{dw}{dt} &= f_x\big(x(t), y(t), z(t)\big)x'(t) + f_y\big(x(t), y(t), z(t)\big)y'(t) \\ &\quad + f_z\big(x(t), y(t), z(t)\big)z'(t).\end{aligned}$$

Example

For $w = e^{x^2}y + y^2\sqrt{z+4}$, where $x = t^3$, $y = \cos t$, and $z = \ln(t^2+1)$, find the value of $\dfrac{dw}{dt}$ at $t = 0$.

Solution. By the Chain Rule, we have

$$\begin{aligned}\frac{dw}{dt} &= \frac{\partial w}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial w}{\partial y}\cdot\frac{dy}{dt} + \frac{\partial w}{\partial z}\cdot\frac{dz}{dt} \\ &= 2xe^{x^2}y\cdot(3t^2) + (e^{x^2} + 2y\sqrt{z+4})\cdot(-\sin t) + \frac{y^2}{2\sqrt{z+4}}\cdot\frac{2t}{t^2+1}.\end{aligned}$$

When $t = 0$, we have $x = 0$, $y = 1$, and $z = 0$. Then,

$$\frac{dw}{dt} = 0 + (1 + 2\sqrt{4})(-\sin 0) + \frac{1}{2\sqrt{4}}\cdot\frac{0}{0+1} = 0.$$

• The Chain Rule - Case 2

- For a function $f(x, y)$, when both x and y are functions of TWO variables.

Suppose $x = g(s, t)$ and $y = h(s, t)$. Then $w = f(x, y)$ is a function of two variables s and t and we may find partial derivatives $\frac{\partial w}{\partial s}$ and $\frac{\partial w}{\partial t}$.

Theorem (Chain Rule for $f(x, y)$: Case 2)

If $w = f(x, y)$, where $x = g(s, t)$ and $y = h(s, t)$ are differentiable, then w has partial derivatives w.r.t. s and t, and

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s},$$

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t}.$$

Example

For $w = e^x \sin y$, where $x = st^2$ and $y = s^2 t$, find $\dfrac{\partial w}{\partial s}$ and $\dfrac{\partial w}{\partial t}$.

Solution. We have

$$\begin{aligned}
\frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s} \\
&= (e^x \sin y) \cdot (t^2) + (e^x \cos y) \cdot (2st) \\
&= t^2 e^{st^2} \sin(s^2 t) + 2ste^{st^2} \cos(s^2 t),
\end{aligned}$$

and

$$\begin{aligned}
\frac{\partial w}{\partial t} &= \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial t} \\
&= (e^x \sin y) \cdot (2st) + (e^x \cos y) \cdot (s^2) \\
&= 2ste^{st^2} \sin(s^2 t) + s^2 e^{st^2} \cos(s^2 t).
\end{aligned}$$

Example

Write down the Chain Rule for the case where $w = f(x, y)$ and $x = x(s, t, u)$, $y = y(s, t, u)$.

Solution. Note that w is a function of s, t, and u.

We have

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s},$$

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t},$$

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u}.$$

Example

Write down the Chain Rule for the case where $w = f(x, y, z)$ and $x = x(s, t)$, $y = y(s, t)$, $z = z(s, t)$.

Solution. Note that w is a function of s and t.

We have

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial s},$$

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t}.$$

Example

For $u = x^4y + y^2z^3$, where $x = rse^t$, $y = rs^2e^{-t}$, and $z = r^2s\sin t$, find the value of $\dfrac{\partial u}{\partial s}$ when $r = 2, s = 1, t = 0$.

Solution. Note that u is a function of x, y, and z.

Since

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial u}{\partial z}\frac{\partial z}{\partial s},$$

we compute

$$\frac{\partial u}{\partial x} = 4x^3y, \quad \frac{\partial u}{\partial y} = x^4 + 2yz^3, \quad \frac{\partial u}{\partial z} = 3y^2z^2,$$

$$\frac{\partial x}{\partial s} = re^t, \quad \frac{\partial y}{\partial s} = 2rse^{-t}, \quad \frac{\partial z}{\partial s} = r^2\sin t.$$

Example (cont.)

When $r = 2, s = 1, t = 0$, we have $x = 2$, $y = 2$, and $z = 0$. Therefore,

$$\frac{\partial u}{\partial x} = 64, \quad \frac{\partial u}{\partial y} = 16, \quad \frac{\partial u}{\partial z} = 0,$$

$$\frac{\partial x}{\partial s} = 2, \quad \frac{\partial y}{\partial s} = 4, \quad \frac{\partial z}{\partial s} = 0,$$

which give

$$\frac{\partial u}{\partial s} = 64 \cdot 2 + 64 \cdot 4 + 0 \cdot 0 = 192.$$

• The General Chain Rule

Theorem (The Chain Rule: General Case)

Let u be a function n variables $x_1, x_2, \ldots, x_n$, each of which is a function of m variables $t_1, t_2, \ldots, t_m$ such that all partial derivatives $\dfrac{\partial x_i}{\partial t_j}$ exist ($i = 1, 2, ..., n$; $j = 1, 2, ..., m$).

Then u is a function of $t_1, t_2, \ldots, t_m$ and

$$\frac{\partial u}{\partial t_j} = \frac{\partial u}{\partial x_1}\frac{\partial x_1}{\partial t_j} + \frac{\partial u}{\partial x_2}\frac{\partial x_2}{\partial t_j} + \cdots + \frac{\partial u}{\partial x_n}\frac{\partial x_n}{\partial t_j}$$

for each $j = 1, 2, ..., m$.

Example

Write down the Chain Rule for the case where $w = f(x, y, z, t)$ and $x = x(u, v)$, $y = y(u, v)$, $z = z(u, v)$, and $t = t(u, v)$.

Solution. We apply the theorem above with $n = 4$ and $m = 2$.

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial u} + \frac{\partial w}{\partial t}\frac{\partial t}{\partial u},$$
$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial v} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial v} + \frac{\partial w}{\partial t}\frac{\partial t}{\partial v}.$$

• Implicit Differentiation

Now suppose that z is defined **implicitly** as a function of x and y by an equation of the form

$$F(x, y, z) = C, \quad \text{where } C \text{ is a constant.}$$

Our aim is to find $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$.

Apply the Chain Rule to differentiate the above equation w.r.t. x

$$\frac{\partial F}{\partial x}\underbrace{\frac{\partial x}{\partial x}}_{1} + \frac{\partial F}{\partial y}\underbrace{\frac{\partial y}{\partial x}}_{0} + \frac{\partial F}{\partial z}\frac{\partial z}{\partial x} = 0$$

($\partial x/\partial x = 1$ and $\partial y/\partial x = 0$, because y is independent to x).

The preceding equation becomes:

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z}\frac{\partial z}{\partial x} = 0.$$

Similarly,

$$\frac{\partial F}{\partial y} + \frac{\partial F}{\partial z}\frac{\partial z}{\partial y} = 0.$$

Suppose that $\dfrac{\partial F}{\partial z} \neq 0$. In this case, we obtain

$$\frac{\partial z}{\partial x} = -\frac{\dfrac{\partial F}{\partial x}}{\dfrac{\partial F}{\partial z}} = \frac{-F_x}{F_z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{\dfrac{\partial F}{\partial y}}{\dfrac{\partial F}{\partial z}} = \frac{-F_y}{F_z}.$$

Example

Find $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$, if $x^3 + y^3 + z^3 + 6xyz = 1$.

Solution. Let $F(x, y, z) = x^3 + y^3 + z^3 + 6xyz$. Then $F(x, y, z) = 1$, and by the above formula, we have

$$\frac{\partial z}{\partial x} = \frac{-F_x}{F_z} = -\frac{3x^2 + 6yz}{3z^2 + 6xy} = -\frac{x^2 + 2yz}{z^2 + 2xy}$$

and

$$\frac{\partial z}{\partial y} = \frac{-F_y}{F_z} = -\frac{3y^2 + 6xz}{3z^2 + 6xy} = -\frac{y^2 + 2xz}{z^2 + 2xy}.$$

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- Recall: Gradient vector.

Recall that for a function $f(x, y)$ of two variables x and y, the gradient of f is

$$\nabla f(x, y) = (f_x, f_y) = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j},$$

whereas for a function $F(x, y, z)$ of three variables x, y, and z, the gradient of F is

$$\nabla F(x, y, z) = (F_x, F_y, F_z) = \frac{\partial F}{\partial x}\mathbf{i} + \frac{\partial F}{\partial y}\mathbf{j} + \frac{\partial F}{\partial z}\mathbf{k}.$$

In this section, we discuss the orthogonal property of the gradient vector to its level curve or level surface.

It is used in describing equation of tangent planes. Then it is applied to estimating errors and linear approximations.

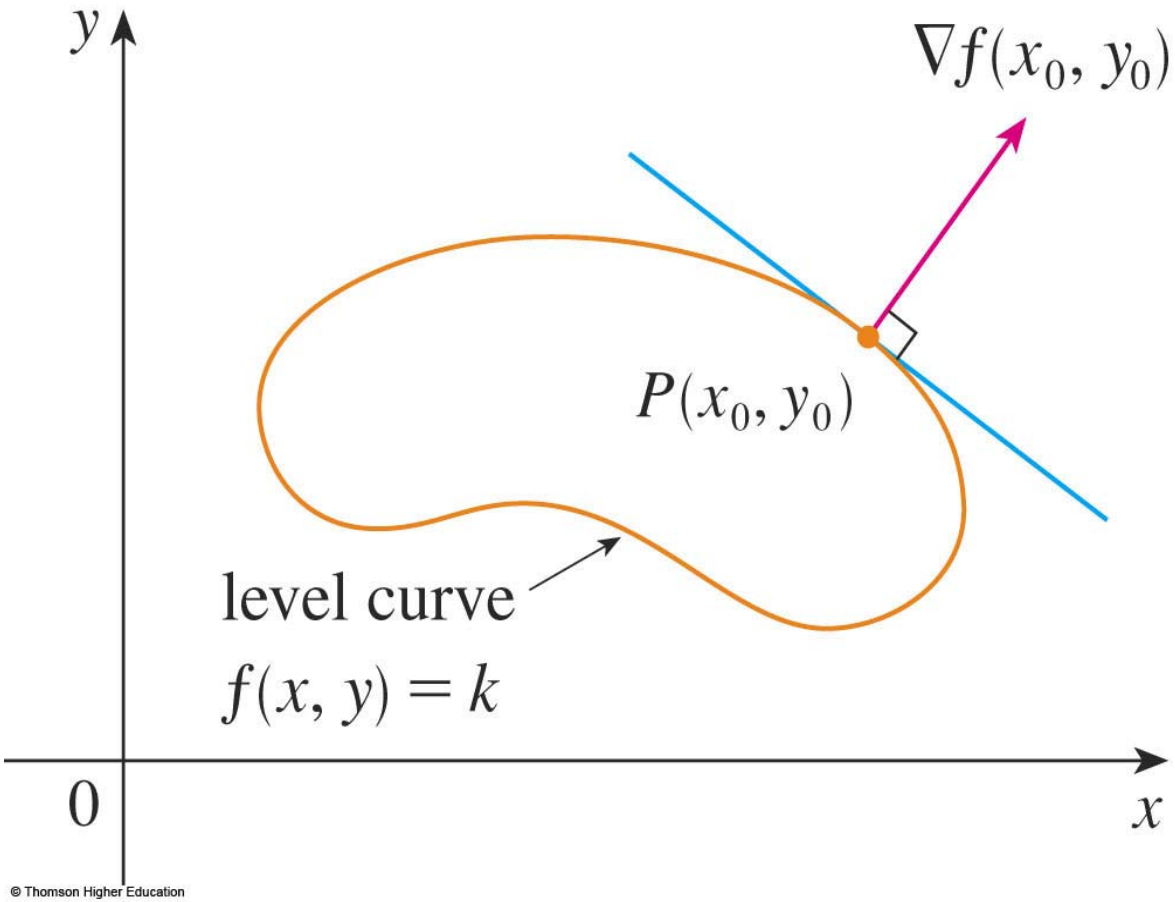
• Gradient vector perpendicular to level curve

Proposition

Consider the 2-variable function $f(x, y)$ and the level curve defined by $f(x, y) = k$. Suppose $P_0(x_0, y_0)$ is a point on this level curve, i.e., $f(x_0, y_0) = k$.

Then the gradient vector $\nabla f(P_0)$ is orthogonal (perpendicular) to the tangent of the level curve $f(x, y) = k$ at P_0.

• Diagram: Orthogonality



Example

Let $f(x, y) = x(x-3) - y$. Sketch the level curve of f which passes through $P_0(1, -1)$ and indicate the gradient vector $\nabla f(P_0)$ on the sketch.

Solution. Note that $f(P_0) = f(1, -1) = -1$.

The level curve passing through P is the curve

$$f(x, y) = -1 \iff x(x-3) - y = -1 \iff y = x^2 - 3x + 1,$$

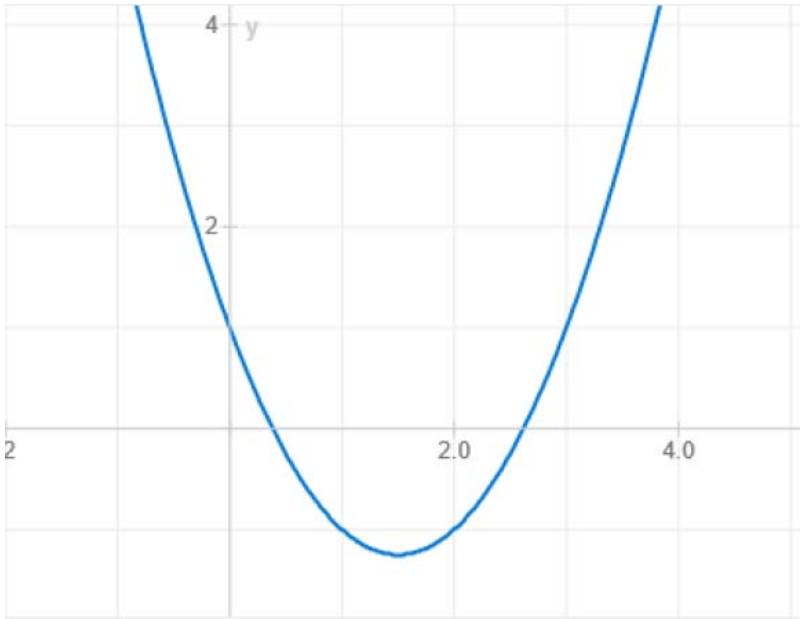
which is the parabola.

The gradient vector is $\nabla f(x, y) = (2x - 3, -1)$, and so

$$\nabla f(P_0) = (-1, -1).$$

At $P_0(1,-1)$, the gradient $\nabla f(P_0) = (-1,-1)$ is perpendicular to the tangent of the parabola $y = x(x-3)+1$.

Sketch of the level curve and the gradient vector $\nabla f(P_0)$:



• Gradient vector perpendicular to level surface

Proposition

Consider a 3-variable function $F(x, y, z)$ and the level surface S is defined by $F(x, y, z) = k$. Suppose $P(x_0, y_0, z_0)$ is a point on S.

Then the gradient vector $\nabla F(P_0)$ is orthogonal (perpendicular) to the tangent of the level surface at P_0.

• Tangent planes to level surfaces

Here with $\nabla F(x_0, y_0, z_0) \neq \mathbf{0}$ (i.e. is not the zero vector), we mean the tangent plane to the level surface $F(x, y, z) = k$ at $P(x_0, y_0, z_0)$ is the plane passes through P and has normal vector $\nabla F(x_0, y_0, z_0)$.

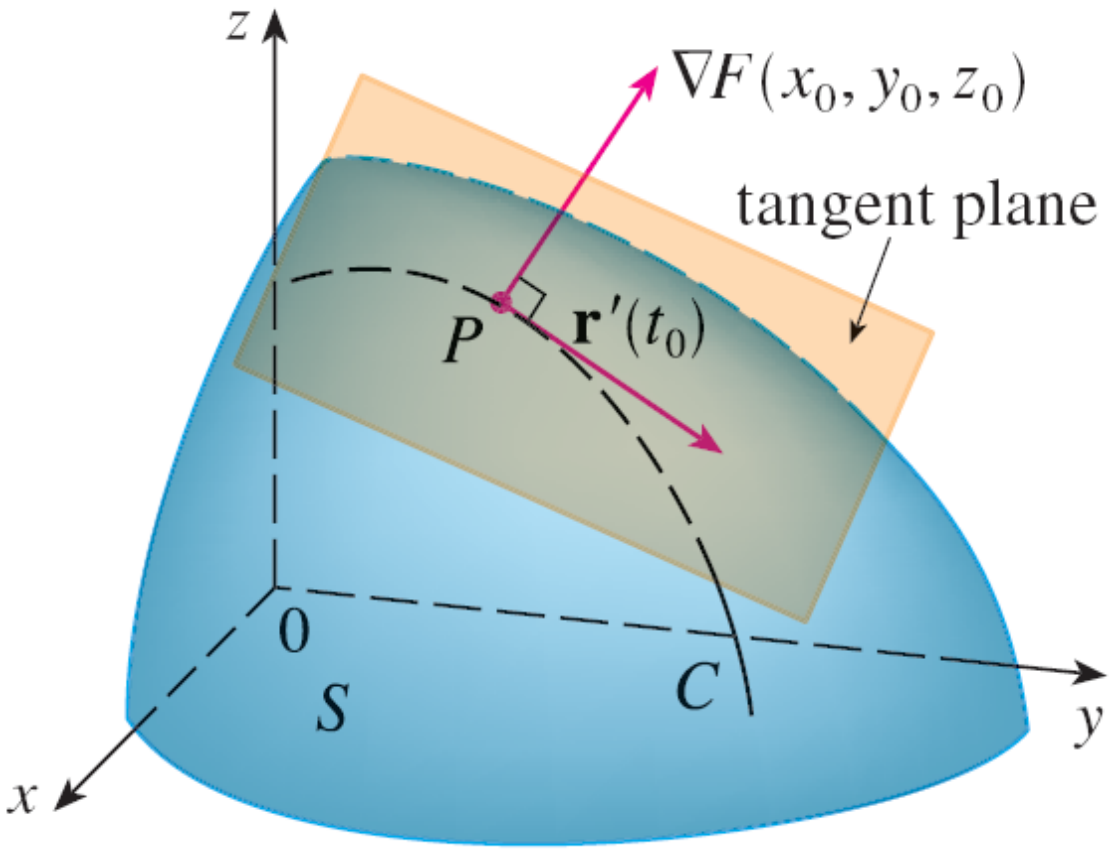
Therefore, the equation of this tangent plane is

$$\nabla F(x_0, y_0, z_0) \cdot (x - x_0, y - y_0, z - z_0) = 0.$$

We can write the equation of this tangent plane as

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0.$$

• Diagram: Orthogonality



• Normal lines to level surfaces

The normal line to the surface $F(x, y, z) = k$ at $P(x_0, y_0, z_0)$ can be represented by the vector equation

$$\mathbf{r} = (x_0, y_0, z_0) + t\nabla F(x_0, y_0, z_0)$$

or

$$\begin{aligned} x &= x_0 + t\, F_x(x_0, y_0, z_0), \\ y &= y_0 + t\, F_y(x_0, y_0, z_0), \\ z &= z_0 + t\, F_z(x_0, y_0, z_0). \end{aligned}$$

Example

Find the equation of the tangent plane and normal at the point $(-2, 1, -3)$ to the ellipsoid

$$\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 3.$$

Solution.

- **Tangent plane.**

Let $F(x, y, z) = \dfrac{x^2}{4} + y^2 + \dfrac{z^2}{9}$. The equation of the tangent plane at (x_0, y_0, z_0) is

$$\nabla F(x_0, y_0, z_0) \cdot (x - x_0, y - y_0, z - z_0) = 0.$$

Example (cont.)

From

$$\nabla F = (F_x, F_y, F_z) = \left(\frac{x}{2}, 2y, \frac{2z}{9}\right),$$

it follows that $\nabla F(-2, 1, -3) = \left(-1, 2, \frac{-2}{3}\right)$.

Hence, the required equation is

$$\begin{aligned}
&\left(-1, 2, \frac{-2}{3}\right) \cdot \big(x - (-2), y - 1, z - (-3)\big) = 0 \\
\iff \quad & -(x+2) + 2(y-1) - \frac{2}{3}(z+3) = 6 \\
\iff \quad & x - 2y + \frac{2}{3}z = -6.
\end{aligned}$$

• **Normal line**.

The equation of the normal line to the ellipsoid $\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 3$ at the point (x_0, y_0, z_0) is

$$\mathbf{r} = (x_0, y_0, z_0) + t\nabla F(x_0, y_0, z_0),$$

and hence, the desired equation is

$$(x, y, z) = (-2, 1, -3) + t(-1, 2, \frac{-2}{3}).$$

This is in a parametric form, that is

$$\begin{cases} x = -2 - t \\ y = 1 + 2t \\ z = -3 - \frac{2}{3}t \end{cases} \qquad t \in \mathbb{R}.$$

• Tangent plane to $z = f(x, y)$

We can obtain an equation of the tangent plane of the graph $z = f(x, y)$, because the latter is viewed as the level surface $F(x, y, z) = 0$, where $F(x, y, z) = f(x, y) - z$.

So we have the following result.

Proposition (Tangent plane to $z = f(x, y)$)

Let $f(x, y)$ be a function of two variables. The equation of the tangent plane to the graph $z = f(x, y)$ at $(x_0, y_0, f(x_0, y_0))$ is given by

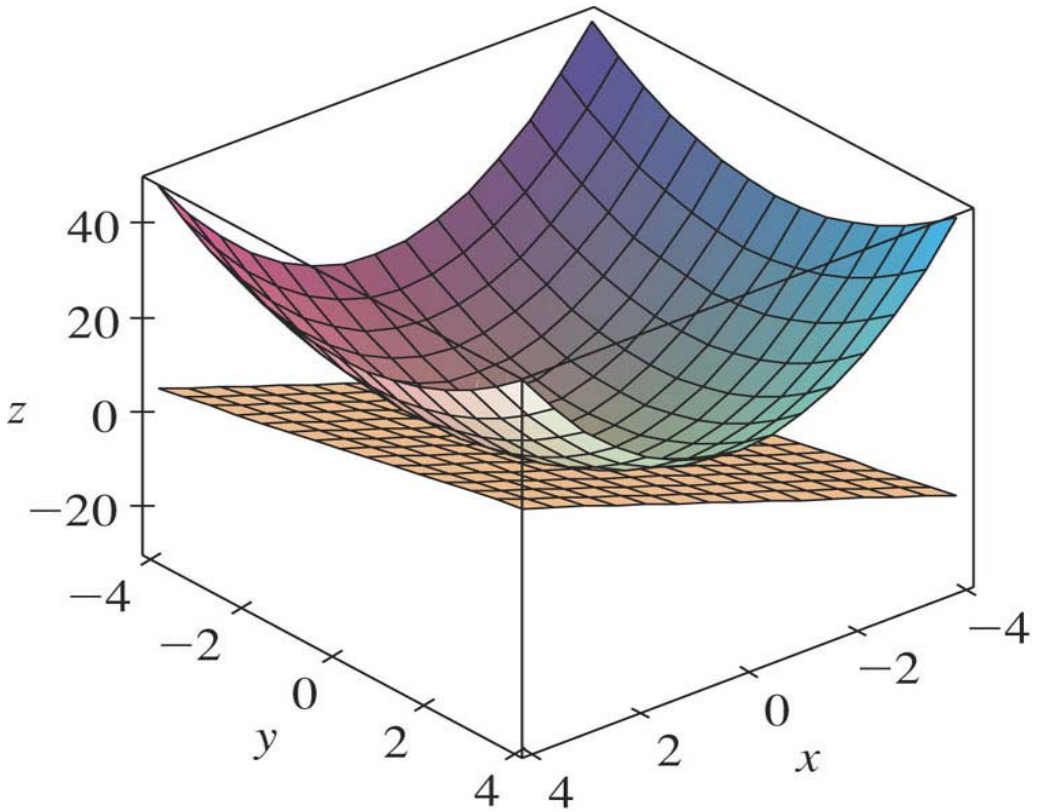
$$z = f(x_0, y_0) + \nabla f(x_0, y_0) \cdot (x - x_0, y - y_0),$$

that is

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

• Diagram: Tangent plane

$$z = f(x_0, y_0) + \nabla f(x_0, y_0) \cdot (x - x_0, y - y_0),$$



(a)

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• Summary: Equation of Tangent Plane to $z = f(x, y)$

We have

- $z = f(x_0, y_0) + \nabla f(x_0, y_0) \cdot (x - x_0, y - y_0)$, or equivalently $z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$.
- Rearranging the above gives

$$\underbrace{f_x(x_0, y_0)}_{a} x + \underbrace{f_y(x_0, y_0)}_{b} y - z$$

$$= \underbrace{f_x(x_0, y_0)x_0 + f_y(x_0, y_0)y_0 - f(x_0, y_0)}_{d},$$

which is in the form $ax + by + cz = d$, where $c = -1$.

• Remark: Tangent plane to $z = f(x, y)$.

- For a function $f(x)$ of one variable, the equation of the tangent line at $x = x_0$ is given by

$$y = f(x_0) + f'(x_0)(x - x_0).$$

- We obtained above a similar equation describing the tangent plane to the surface $z = f(x, y)$ at $(x_0, y_0, f(x_0, y_0))$ given by

$$z = f(x_0, y_0) + \nabla f(x_0, y_0) \cdot (x - x_0, y - y_0),$$

that is,

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Example

Find the equation of the tangent plane to the surface $z = 2x^2 + y^2$ at the point $(1, 1, 3)$.

Solution. Let $f(x, y) = 2x^2 + y^2$. The required equation is

$$z = f(1, 1) + f_x(1, 1)(x - 1) + f_y(1, 1)(y - 1).$$

Note that $f_x(x, y) = 4x$ and $f_y(x, y) = 2y$. So we obtain $f(1, 1) = 3$, $f_x(1, 1) = 4$, and $f_y(1, 1) = 2$.

Therefore, the equation of the tangent plane at $(1, 1, 3)$ is

$$z = 3 + 4(x - 1) + 2(y - 1), \quad \text{i.e.} \quad z = 4x + 2y - 3.$$

• Recall (Calculus I): Linear Approximation of $f(x)$

Recall that a linear approximation of a single variable function $f(x)$ for values of x near a is an approximation of $f(x)$ using the tangent function.

(Note that $y = f(a) + f'(a)(x - a)$ is the equation of tangent line to $y = f(x)$ at $x = a$.)

The value $f(x)$ is approximated by the function $L(x)$ which defines the tangent line:

$$f(x) \approx L(x) = f(a) + f'(a)(x - a).$$

• Linear approximation of $f(x, y)$

For a 2-variable function $f(x, y)$, we have a similar linear approximation for (x, y) close to (a, b).

Recall that the equation of tangent plane to $z = f(x, y)$ at (a, b) is

$$z = \underbrace{f(a, b) + f_x(a, b) \cdot (x - a) + f_y(a, b) \cdot (y - b)}_{L(x,y)}.$$

Let

$$L(x, y) = f(a, b) + f_x(a, b) \cdot (x - a) + f_y(a, b) \cdot (y - b)$$

be the function which describes the above tangent plane.

Proposition (Linear approximation of $f(x, y)$)

For (x, y) close to (a, b), we use $L(x, y)$ to approximate $f(x, y)$ as follows:

$$f(x, y) \approx L(x, y),$$

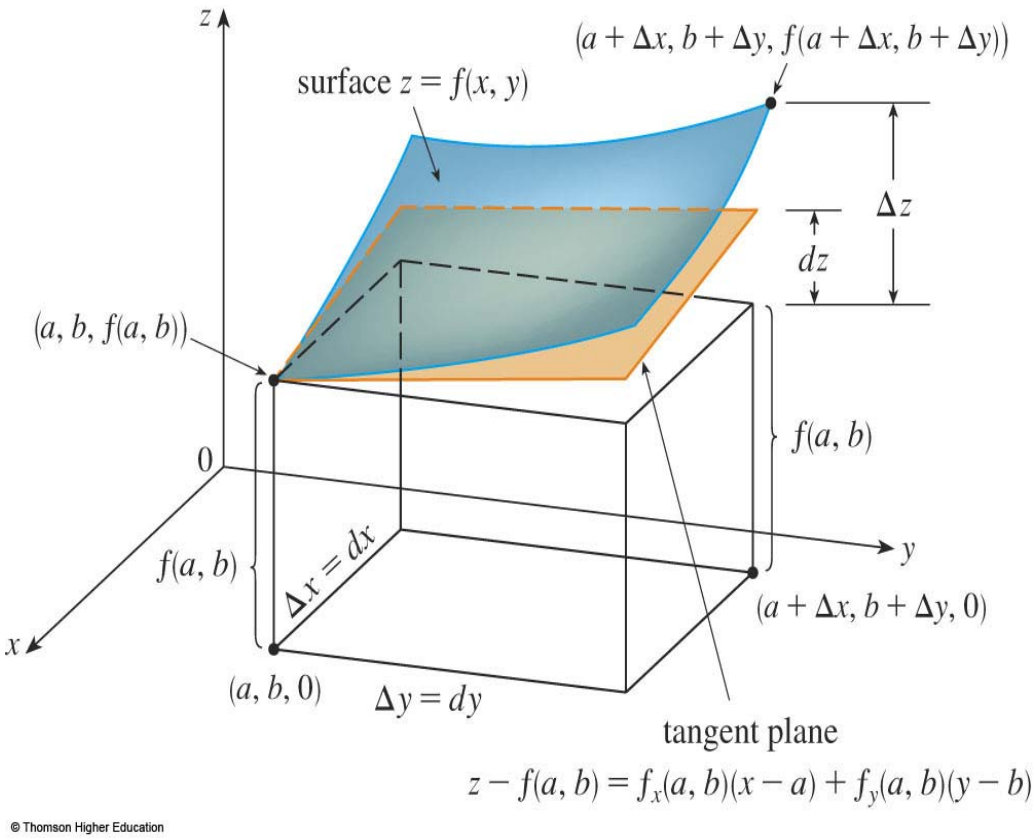
that is

$$f(x, y) \approx f(a, b) + f_x(a, b) \cdot (x - a) + f_y(a, b) \cdot (y - b)$$

$$= f(a, b) + \nabla f(a, b) \cdot (x - a, y - b).$$

• Diagram: Linearization

$$f(x, y) \approx L(x, y) = f(a, b) + f_x(a, b) \cdot (x - a) + f_y(a, b) \cdot (y - b)$$



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Example

Use linear approximation to approximate the value

$$\sqrt{9(1.95)^2 + (8.1)^2}.$$

Solution. Let $f(x, y) = \sqrt{9x^2 + y^2}$.

Note that 1.95 is close to 2 and 8.1 is close to 8.

Then we choose $a = 2$ and $b = 8$ to approximate $f(1.95, 8.1)$:

$$f(x, y) \approx L(x, y) = f(a, b) + f_x(a, b) \cdot (x - a) + f_y(a, b) \cdot (y - b)$$

which gives

$$f(1.95, 8.1) \approx f(2, 8) + f_x(2, 8) \cdot (1.95 - 2) + f_y(2, 8) \cdot (8.1 - 8).$$

Example (cont.)

We have

$$f_x(x,y) = \frac{\partial}{\partial x}\sqrt{9x^2+y^2} = \frac{9x}{\sqrt{9x^2+y^2}},$$
$$f_y(x,y) = \frac{\partial}{\partial y}\sqrt{9x^2+y^2} = \frac{y}{\sqrt{9x^2+y^2}}.$$

Then we have

$$\begin{aligned} f(1.95, 8.1) &\approx f(2,8) + f_x(2,8)\cdot(-0.05) + f_y(2,8)\cdot(0.1) \\ &= 10 + \frac{9\cdot 2}{\sqrt{9\cdot 2^2+8^2}}(-0.05) + \frac{8}{\sqrt{9\cdot 2^2+8^2}}(0.1) \\ &= 10 + (-0.01) = 9.99. \end{aligned}$$

• Estimate of change in f: Differentials

Let $f(x, y)$ be a function of two variables. We study how to estimate the change in f from a point (a, b) to $(a + \Delta x, b + \Delta y)$, i.e.

$$\Delta f = f(a + \Delta x, b + \Delta y) - f(a, b).$$

Here, Δx and Δy are respectively small changes in variables x and y.

- **Recall (Calculus I):** for a function $f(x)$ of one variable, to estimate the change $\Delta f = f(a + \Delta x) - f(a)$, we use $L(a + \Delta x)$ for $f(a + \Delta x)$, where $L(x) = f(a) + f'(a)(x - a)$ is the linear approximation of f about a.

Thus we have

$$\Delta f \approx df = L(a + \Delta x) - f(a) = f'(a)\Delta x,$$

where $df = f'(a)\Delta x$ is known as the differential of f.

• Change & Total Differential

- Similarly, when Δx and Δy are small, to estimate Δf, we use $L(a+\Delta x, b+\Delta y)$ for $f(a+\Delta x, b+\Delta y)$.

Recall: $L(x,y) = f(a,b) + \nabla f(a,b) \cdot (x-a, y-b)$.

Thus we have

$$\begin{aligned} \Delta f &= f(a+\Delta x, b+\Delta y) - f(a,b) \\ &\approx L(a+\Delta x, b+\Delta y) - f(a,b) \\ &= \nabla f(a,b) \cdot (\Delta x, \Delta y) \\ &= f_x(a,b)\Delta x + f_y(a,b)\Delta y, \end{aligned}$$

• Increment & Total Differential

Definition

We define the total differential of f (or simply, the differential of f), denoted by df, as

$$df = \nabla f(a,b) \cdot (\Delta x, \Delta y) = f_x(a,b)\Delta x + f_y(a,b)\Delta y.$$

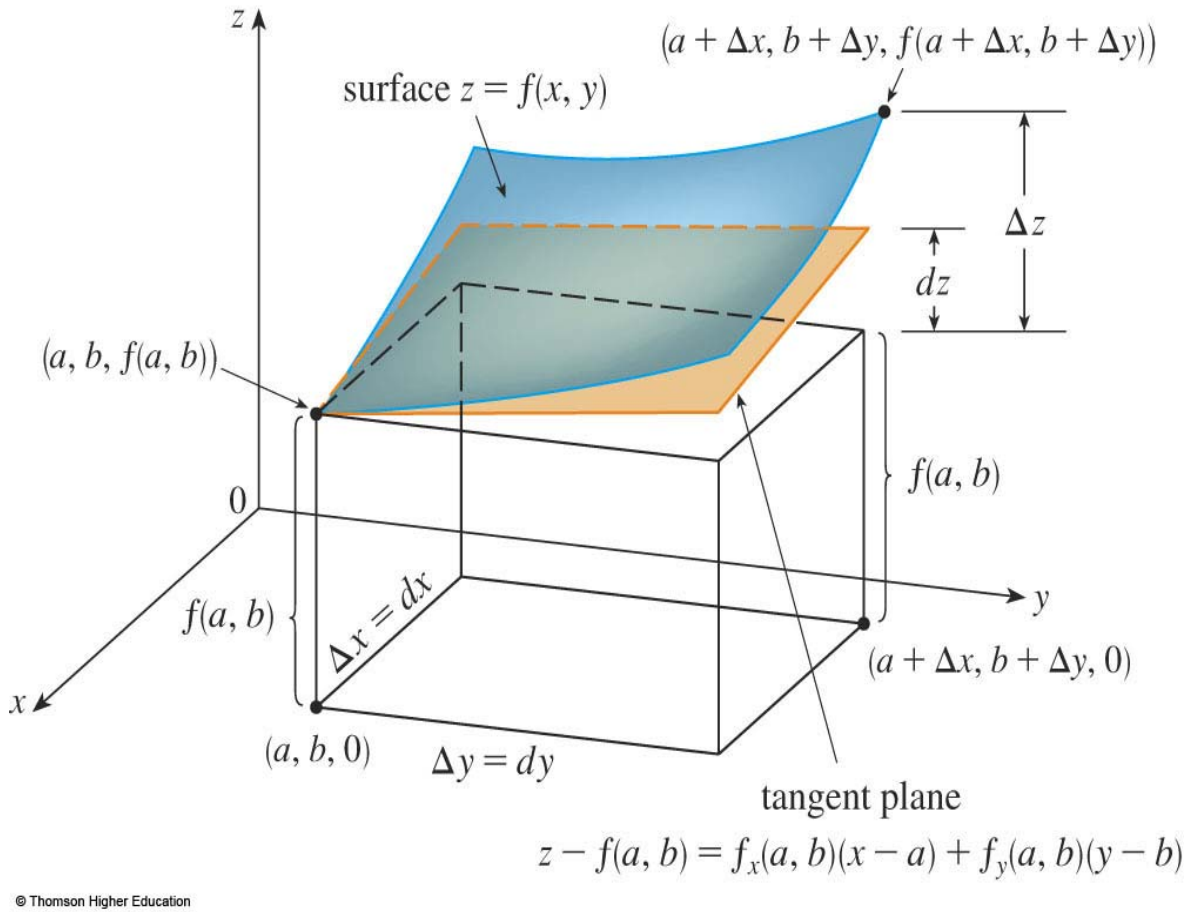
Thus we have $\Delta f \approx df$.

Remark. For independent variables x and y, the following notation is often used: $dx = \Delta x$ and $dy = \Delta y$, so that

$$df = \nabla f(a,b) \cdot (dx, dy) = f_x(a,b)dx + f_y(a,b)dy.$$

• Diagram: Differential $z = f(x, y)$

We have $\Delta z \approx dz = f_x(a, b)dx + f_y(a, b)dy$.



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Example

Let $f(x,y) = x^2 + 3xy - y^2$. If x changes from 2 to 2.05 and y changes from 3 to 2.96, compute df and compare it with Δf.

Solution. • Since x changes from 2 to 2.05 and y changes from 3 to 2.96, we have

$$a = 2 \implies \Delta x = 2.05 - 2 = 0.05$$
$$b = 3 \implies \Delta y = 2.96 - 3 = -0.04.$$

Also

$$f_x(x,y) = 2x + 3y \quad \text{and} \quad f_y(x,y) = 3x - 2y.$$

Then we have

$$\begin{aligned} df &= (2a+3b)\Delta x + (3a-2b)\Delta y \\ &= (2\cdot 2 + 3\cdot 3)\cdot(0.05) + (3\cdot 2 - 2\cdot 3)\cdot(-0.04) = 0.65. \end{aligned}$$

- On the other hand, the change in f is

$$\begin{aligned}
\Delta f &= f(a+\Delta x, b+\Delta y) - f(a,b) \\
&= f(2.05, 2.96) - f(2,3) \\
&= \left[(2.05)^2 + 3(2.05)\cdot(2.96) - (2.96)^2\right] - \left[(2)^2 + 3(2)\cdot(3) - (3)^2\right] \\
&= 0.6449.
\end{aligned}$$

Note that Δf is approximately equal to df, but df is much easier to calculate.

• Functions of $n \geq 3$ variables

Definition

For a function $F(x, y, z)$ of three variables, we have the following linear approximation and total differential at (a, b, c) as follows.

- Linear approximation:

$$\begin{aligned} F(x, y, z) &\approx L(x, y, z) \\ &= F(a, b, c) + \nabla F(a, b, c) \cdot (x - a, y - b, z - c), \end{aligned}$$

- Total differential:

$$\Delta F \approx df = \nabla F(x, y, z) \cdot (x - a, y - b, z - c)$$

$$= F_x(a, b, c) \cdot (x - a) + F_y(a, b, c) \cdot (y - b) + F_z(a, b, c) \cdot (z - c).$$

More generally, for a function $f(x_1, x_2, \ldots, x_n)$ of n variables, we have the following linear approximation and total differential at $(a_1, a_2, \ldots, a_n)$.

Definition

- Linear approximation:

$$\begin{aligned} f(x_1, x_2, \ldots, x_n) &\approx L(x_1, x_2, \ldots, x_n) \\ &= f(a_1, a_2, \ldots, a_n) \\ &+ \nabla f(a_1, a_2, \ldots, a_n) \cdot (x_1 - a_1, x_2 - a_2, \ldots, x_n - a_n), \end{aligned}$$

- Total differential:

$$df = \nabla f(a_1, a_2, \ldots, a_n) \cdot (x_1 - a_1, x_2 - a_2, \ldots, x_n - a_n).$$

END OF CHAPTER 1